Native nation API

The native nation route combines fifteen same-version domestic city replies and available nationwide air rows. It shares the existing S3 inputs and provider acquisitions; it never forwards a selected nation request to legacy.

Setup

Use the existing server2 configuration (server2/.env). Add kma to SERVER2_NATIVE_FAMILIES for /v000903/nation/KR, or kma-v000901 for /v000901/nation/KR. These are the existing version switches, not new modes. Provider keys keep their legacy names. S3 remains authoritative; deleting memory or an optional local cache must not affect correctness. This document describes implemented routes and local verification, not a production deployment.

Client behavior

A nation request starts all fifteen fixed-city assemblies with one absolute three-second deadline. Each source returns the data available for that request; late acquisition continues independently and is shared by later requests. A missing optional provider branch does not by itself turn the response into an error. If an actual city dependency fails, the nation route returns the legacy rendered 500 error. Successful child replies produce 200.

Province air prefers fresh urban-station means. Otherwise it uses a labelled representative city point and reports its source and safe reason in airStatus. coverage: "representative-point" must not be described as a province mean. The national provider query is shared with ordinary domestic weather.

Verified local scenarios

The actual loopback HTTP handler was called with both versions, GET, HEAD and If-None-Match. The test legacy origin was a closed port. All responses remained native: GET 200, HEAD 200 with no body, and revalidation 304 with no body.

The nation-specific tests also publish raw records to a loopback S3 peer and reconstruct all seventeen province means twice using new empty-memory sources with providers disabled. Twelve concurrent requests share one intentionally slow national acquisition. The first requests return available data within their three-second budget; the next request includes the newly saved observation.

```
$ cargo test --locked --manifest-path server2/Cargo.toml --test nation
Actual local HTTP: v000903 and v000901 GET=200 (15 cities, 17 provinces), HEAD=200 (empty), ETag
revalidation=304 (empty); legacy origin closed.
test result: ok. 10 passed; 0 failed; 0 ignored
```

Run the maintained checks from the repository root:

```
cargo test --locked --manifest-path server2/Cargo.toml --test nation
cargo clippy --locked --manifest-path server2/Cargo.toml --test nation -- -D warnings
```

For the offline legacy oracle use pinned Node 16.20.2, Mongoose 5.1.2 and the legacy async/Jade dependencies. nation_record.cjs takes repository and output paths. It does not load private configuration or contact providers or databases. Its city weather dependencies are deterministic doubles; it verifies nation composition, not live weather-provider output.

Evidence limits

No additional live provider request is claimed: this slice reuses the already verified national AirKorea method. No AWS IAM/TLS behavior, deployed latency, cutover, device push, or operational quota is proven by a loopback test. Native diagnostic error HTML omits the legacy session ID because server2 does not create legacy session cookies. Exact source and oracle boundaries are in server2/tests/nation/PROVENANCE.md.